

Advanced Praktika SoSe 26 Notes

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Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

Part I

Notes

1 Project Setup

1.1 General repository structure

The whole project is managed inside a Quarto book repository. The Quarto project contains notes, theory summaries, the final report, and the implementation code.

The planned directory structure is:

```
project-root/  
  index.qmd  
  report/  
  notes/  
  code/  
    CMakeLists.txt  
    external/  
      hdnum/          # git submodule  
        cpfloat/      # cloned inside hdnum  
    include/  
      mixed_ir.hpp    # algorithm  
      hdnum_conversions.hpp # convert()  
      error_metrics.hpp # forward/backward error helpers, later  
    experiments/  
      exp_convergence.cc  
      exp_condition_sweep.cc  
      exp_residual_precision.cc  
    examples/  
      hdnum_sandbox.cc  
    tests/  
      test_conversion.cc  
      test_mixed_ir_sanity.cc  
    scripts/  
      run_all_experiments.py  
      plot_results.py  
    results/  
      raw/  
      plots/
```

Purpose of the main sub-directories:

external/	external dependencies, especially HDNUM
include/	own header-only algorithm and helper code
src/	optional implementation files, if needed later
examples/	small sandbox/demo programs for learning and testing
experiments/	reproducible experimental drivers for report data
tests/	correctness and sanity checks
scripts/	automation and plotting scripts
results/raw/	generated CSV or data files
results/plots/	generated figures for the report

1.1.1 Experimental Drivers

They will go in:

```
code/  
  experiments/  
    exp_convergence.cc  
    exp_condition_sweep.cc  
    exp_residual_precision.cc
```

They will be written in c++ because they need to instantiate HDNUM types like FP16, bfloat16, FP64, FP128, FP256, run LU, perform refinement, and compute errors.

But the overall experiment workflow will be split:

```
C++ drivers:  
  run numerical experiments  
  write CSV files  
  
Python / shell scripts:  
  run many driver configurations  
  collect results  
  make plots
```

like:

```
code/
  experiments/
    exp_convergence.cc
    exp_condition_sweep.cc
    exp_residual_precision.cc
  scripts/
    run_all_experiments.py
    plot_results.py
  results/
    raw/
    plots/
```

The cpp experiments can be written so that they can be run like:

```
./build/exp_condition_sweep --uf FP16 --u FP64 --ur FP128 --n 100 --matrix random_orthogonal
./build/exp_condition_sweep --uf bfloat16 --u FP64 --ur FP128 --n 100 --matrix random_orthogonal
./build/exp_condition_sweep --uf FP32 --u FP64 --ur FP128 --n 100 --matrix random_orthogonal
```

They a python or shell script can run all of those automatically.

Precision sweeps should vary precision, condition number, and matrix type, and experiments should be scripted so rerunning them is convenient. In practice, that means your experiment system should let you rerun a whole table of experiments with one command, rather than manually recompiling/editing files.

1.2 HDNUM, CPFloat and GMP

HDNUM is included as a git submodule:

```
git submodule add https://parcomp-git.iwr.uni-heidelberg.de/Teaching/hdnum.git code/external/hdnum
git submodule update --init --recursive
```

CPFloat is cloned and built inside the HDNUM top-level directory, matching HDNUM's expected layout:

```
cd code/external/hdnum
git clone https://github.com/north-numerical-computing/cpfloat.git
cd cpfloat
make lib
```


GMP is installed system-wide:

```
sudo apt install libgmp-dev
```

Because CPFloat sits inside the HDNUM submodule but is not part of HDNUM itself, untracked files are hidden locally only for that submodule:

```
cd code/external/hdnum
git config status.showUntrackedFiles no
```

and the parent repository ignores untracked content specifically inside the HDNUM submodule:

```
git config submodule.code/external/hdnum.ignore untracked
```

1.3 Changing Top-level CMakeLists.txt

Currently the implementations are header-only and are contained in the folder include. Later if source files that not only header are added e.g. to a src/ folder it has to be accounted for:

Right now, this part:

```
add_library(mixed_precision_core INTERFACE)
```

is only an INTERFACE target. That means it does not compile any .cc files. It only carries usage requirements:

```
include_directories
compile_definitions
linked_libraries
compile_features
```

So it works well for a header-only project, but it does not yet build files from src/.

But if later you add:

```
src/csv_writer.cc
src/matrix_generators.cc
src/experiment_io.cc
```

then those files will not automatically be compiled unless we update CMake.

1.3.1 How to integrate src/ later

The usual clean solution is to change from a purely INTERFACE core target to a real library target.

For example:

```
add_library(mixed_precision_core
    src/csv_writer.cc
    src/matrix_generators.cc
    src/experiment_io.cc
)
```

Then keep the same include directories and dependencies:

```
target_include_directories(mixed_precision_core
    PUBLIC
        "${CMAKE_CURRENT_SOURCE_DIR}/include"
        "${HDNUM_DIR}"
        "${CPFLOAT_INCLUDE_DIR}"
)

target_compile_definitions(mixed_precision_core
    PUBLIC
        HDNUM_HAS_GMP=1
        HDNUM_HAS_CPFLOAT=1
)

target_link_libraries(mixed_precision_core
    PUBLIC
        "${CPFLOAT_LIBRARY}"
        gmpxx
        gmp
)

target_compile_features(mixed_precision_core
    PUBLIC
        cxx_std_17
)
```

Then all your examples, tests, and experiments can continue doing:

```
target_link_libraries(my_target
    PRIVATE
        mixed_precision_core
        mixed_precision_warnings
)
```

and they will automatically get both:

```
the headers from include/
the compiled implementation files from src/
```

1.3.2 A flexible version

If you do not want to manually list every source file, you can use globbing:

```
file(GLOB project_sources CONFIGURE_DEPENDS
    "${CMAKE_CURRENT_SOURCE_DIR}/src/*.cc"
    "${CMAKE_CURRENT_SOURCE_DIR}/src/*.cpp"
)

add_library(mixed_precision_core
    ${project_sources}
)
```

But there is one small issue: `add_library()` with an empty source list can be awkward in some CMake versions/workflows.

So for now, while `src/` is empty, I would keep the current `INTERFACE` version.

Later, when you add the first real `.cc` file, change:

```
add_library(mixed_precision_core INTERFACE)
```

to:

```
file(GLOB project_sources CONFIGURE_DEPENDS
    "${CMAKE_CURRENT_SOURCE_DIR}/src/*.cc"
    "${CMAKE_CURRENT_SOURCE_DIR}/src/*.cpp"
)

add_library(mixed_precision_core
```

```
    ${project_sources}  
)
```

and replace INTERFACE with PUBLIC in the relevant `target_*` commands.

1.3.3 Rule of thumb

Use this now:

```
add_library(mixed_precision_core INTERFACE)
```

while the project is header-only.

Use this later:

```
add_library(mixed_precision_core ${project_sources})
```

once `src/` contains real compiled support code.

2 Project Summary

2.1 Abstract

This *Fortgeschrittenenpraktikum* project investigates mixed precision iterative refinement for solving linear systems

$$Ax = b.$$

The goal is to implement a three-precision LU-based method: the LU factorization and correction solves use low precision u_f , the solution updates use working precision u , and the residuals are computed in higher precision u_r . Starting from a low-precision LU solve, the algorithm repeatedly computes $r_i = b - Ax_i$, solves for a correction d_i , and updates x_i until $|d_i|/|x_i| < u$ or a maximum iteration count is reached.

The focus is empirical accuracy, not runtime, since low precisions such as fp8, fp16, and bfloat16 are simulated via CPFloat. The experiments will study convergence, forward/backward error, condition-number dependence, precision triples (u_f, u, u_r) residual precision, and comparison with a direct working-precision solve.

Summary:

- implement three-precision variant of **iterative refinement for linear systems**
- study its accuracy

2.1.1 Problem

- A linear system $Ax = b$. **iterative refinement** improves an approximate solution by repeatedly computing the residual and solving a correction equation.
 - LU factorization in low precision: u_f
 - **working arithmetic** in ‘middle’ precision: u
 - residual computation in high precision: u_r

i Note

Review the definition of **accuracy** of an algorithm

- precision triples (u_r, u, u_f) with $u_f \succ u \succ u_r$:
 - i.e. u_f low precisions like: `fp8`, `fp16`, `bfloat16`, come from `CPFloat`:
 - * `FP8`: `CPFloat<4, 4>`
 - * `bfloat16`: `CPFloat<8, 8>`
 - * `FP16`: `CPFloat<11, 5>`
 - u is probably the native `FP64` i.e. double precision

i Note

Study the `FP8`, `FP16` and `bfloat16` formats and how they relate to template parameters in `CPfloat<>`

2.1.2 Algorithm

Algorithm: Three-precision iterative refinement

1. Compute the LU factorization $PA = LU$ in precision u_f .
2. Solve $LU x_0 = P b$ in precision u_f ; cast x_0 to precision u .
3. While not converged:
 4. Compute $r_i = b - A x_i$ in precision u_r .
 5. Cast r_i to precision u .
 6. Solve $LU d_i = P r_i$ in precision u_f ; cast d_i to precision u .
 7. Update $x_{i+1} = x_i + d_i$ in precision u .

- convergence condition: $\frac{\|d_i\|}{\|x_i\|} < u$ where u is the unit roundoff of the working precision,
- or stop after a max number of iterations.

i Note

why is $\frac{\|d_i\|}{\|x_i\|} < u$ the convergence condition?

2.1.3 HDNUM & Own Implementation

relevant parts:

matrix and vector classes:

- `src/densematrix.hh`: `DenseMatrix<T>` with:
 - `mv`: matrix vector product
 - `mm`: matrix matrix product
 - `transpose()` etc
- `src/vector.hh`: `Vector<T>` with vector arithmetic operations (multiplication with scalar, addition) and norms
- LU decomposition in `src/lr.hh`:
 - `lr_fullpivot(A, p, q)`
 - `lr_partialpivot(A, p)`
 - triangular solves: `solveL(A, x, b)`, `solveR(A, x, b)`
 - `permute_forward` and `permute_backward`

2.1.4 Mixing Types

In a mixed precision algorithm different parts of the algorithm are instantiated / computed with different precisions, i.e. different types:

- LU factorization is done in lower precision u_f : E.g. `DenseMatrix<FP16>`
- Residual computation in higher precision u_r : E.g. `DenseMatrix<FP128>`

Therefore we must be able to convert from one type (class) like `DenseMatrix<FP16>` to another type (class) like `DenseMatrix<FP128>`. This conversion requires explicit construction or element-wise copying.

Managing type conversions between precisions is they key challenge. It is suggested to do this with a explicit template functions for matrices and vectors:

```
template<class T_out, class T_in>
void convert(Vector<T_out>& out, const Vector<T_in>& in)

template<class T_out, class T_in>
void convert(DenseMatrix<T_out>& out, const Vector<T_in>& in)
```

2.1.5 Methodology

- **reference solutions**: run a standard solver for $A \cdot x = b$ in FP256 and treat the result x_{ex} as exact.
- **error metrics**:
 - report errors using norms appropriate to your algorithm

- compute **all error measures** in a precision higher than the working precision of your algorithm, i.e. u
- otherwise you are measuring the precision of your measurement.

i Note

- What does **error** mean in this context? $\|x^* - x_{ex}\|_2$?
- What is a norm that is **appropriate** to LU, simply $\|\bullet\|_2$?
- Does Compute error measures in a precision higher than u simply mean that we compute $\|\cdot\|$ in something like FP128
 - what does “otherwise you are measuring the precision of your measurement” mean?

- experiments should be scripted so that re-running them with various precision, condition number and matrix type is convenient.

2.1.6 What Should be Studied

- **Convergence Histories:** for each precision combination (u_f, u, u_r) and several test matrices A_i with different condition numbers $k(A_i)$:
 - plot **forward error** as a function of i
 - plot **backward error** as a function of i

i Note

review the definitions of **forward error** and **backward error**

- **Summary Across Condition Numbers:** for a wide range of $k(A)$ for each precision combinations
 - record:
 - * the final achieved **forward error**
 - * number of iterations
 - plot these as a function of $k(A)$: theory predicts convergence requires roughly $k(A) \cdot u_f < 1$.
- **The Role u_r :** Fix (u_f, u) and vary u_r . When does $u_r > u$ make a measurable difference

i Note

- Normally $u_r < u$, i.e. the residual computation is done in a precision higher than the working precision.
- What happens when we lower the precision, when does it make a difference?

2.1.7 Plan

1. Implement `hdnum_conversions.hpp`.
2. Add conversion tests.
3. Implement a minimal `mixed_ir.hpp`.
4. First test with all precisions set to FP64.
5. Then test real mixed precision triples.
6. Add systematic experiments and plotting scripts.

3 Conversions

3.1 Floating Point Types

In this project there are three groups of number formats:

1. simulated low precision via CPFloat
2. native C++ hardware types `float` and `double`
3. high / arbitrary precision via GMP

The low precision `fp8`, `fp16`, `bfloat16` are simulated via CPFloat.

An overview of types defined in HDNUM:

HDNUM / project type	Provider	Parameters	Corresponds to in Higham–Mary	Unit roundoff, roughly
FP8	CPFloat	CPFloat<4,4>	custom 8-bit / “quarter precision”-like format	$2^{-4} \approx 6.25 \cdot 10^{-2}$
bfloat16	CPFloat	CPFloat<8,8>	Higham–Mary <code>bfloat16</code>	$2^{-8} \approx 3.91 \cdot 10^{-3}$
FP16	CPFloat	CPFloat<11,5>	IEEE half precision, Higham–Mary <code>fp16</code>	$2^{-11} \approx 4.88 \cdot 10^{-4}$
FP32	native C++ <code>float</code>	effectively (24,8)	IEEE single precision, Higham–Mary <code>fp32</code>	$2^{-24} \approx 5.96 \cdot 10^{-8}$
FP64	native C++ <code>double</code>	effectively (53,11)	IEEE double precision, Higham–Mary <code>fp64</code>	$2^{-53} \approx 1.11 \cdot 10^{-16}$
FP128	GMP	approx. 128-bit, FP<64>	software high precision / “quadruple-like” role, not necessarily IEEE <code>fp128</code>	depends on HDNUM wrapper
FP256	GMP	approx. 256-bit, FP<192>	multiprecision reference precision	depends on chosen GMP precision

HDNUM / project type	Provider	Parameters	Corresponds to in Higham–Mary	Unit roundoff, roughly
FP512, FP1024	GMP	approx. 512 / 1024-bit	multiprecision, beyond Higham–Mary’s standard table	depends on chosen GMP precision

As a reminder the three-precision LU-IR algorithm is:

1. Compute the LU factorization $PA = LU$ in precision u_f .
2. Solve $LU x_0 = P b$ in precision u_f ; cast x_0 to precision u .
3. While not converged:
 4. Compute $r_i = b - A x_i$ in precision u_r .
 5. Cast r_i to precision u .
 6. Solve $LU d_i = P r_i$ in precision u_f ; cast d_i to precision u .
 7. Update $x_{i+1} = x_i + d_i$ in precision u .

i.e.:

- 1) Compute LU in u_f
- 2) Solve the initial system in u_f
- 3) Compute residuals in u_r
- 4) Cast residuals to u
- 5) Solve correction equations using the low-precision LU factors in u_f
- 6) update in u

For the algorithm the intended mapping is:

Algorithm role	Meaning	Typical project choices
u_f	factorization precision; low precision used for LU	FP8, bfloat16, FP16, sometimes FP32
u	working precision; stores A, b, x_i and updates solution	usually FP32 or FP64
u_r	residual precision; computes $r_i = b - Ax_i$ accurately	FP64, FP128, maybe FP256 for experiments

A reasonable set of experiments could be:

Experiment	u_f	u	u_r
baseline sanity check	FP64	FP64	FP64
classical mixed precision	FP32	FP64	FP64
half/double refinement	FP16	FP64	FP64
half/double with higher residual	FP16	FP64	FP128
bfloat comparison	bfloat16	FP64	FP64
very low precision stress test	FP8	FP64	FP128

3.2 Type Conversions

Again, the precisions for the algorithm are:

- 1) u_f : factorization / solve precision (low)
- 2) u : working precision (medium)
- 3) u_r : residual precision (high)

LU is computed in u_f , the initial solution is cast to u , the residual is computed in u_r and cast back to u , the correction solve uses the low-precision LU factors, and the correction is cast to u before updating.

So the practically necessary conversions are:

Conversion		
$u \rightarrow u_f$		create low-precision copies of A , b , or residuals for LU / triangular solves.
$u_f \rightarrow u$		bring x_0 and correction d_i back to working precision.
$u \rightarrow u_r$		compute $r_i = b - Ax_i$ in high precision.
$u_r \rightarrow u$		The residual is cast back before solving the correction equation.
$u_f \leftrightarrow u_r$	not strictly needed	You normally convert through the working representation, or independently convert from the original A, b, x .

3.2.1 Implementing Conversions - Naive Approach

HDNum implements wrappers `FP` and `CPFloat` for the high-precision GMP and the low precision `CPFloat` types in the source files `highprec_gmp.hh` and `lowprec_cpfloat.hh`.

We are asked not to modify HDNum source files, instead implement explicit conversion functions:

```

template<class T_out, class T_in>
void convert(Vector<T_out>& out, const Vector<T_in>& in);

template<class T_out, class T_in>
void convert(DenseMatrix<T_out>& out, const DenseMatrix<T_in>& in);

```

The approach is to assign to `out` element-wise in loops, where the values constructed from the corresponding elements of `in`. A naive approach is to do:

```

#pragma once

#include <cstdlib>
#include "hdnum.hh"

namespace mpir {

// Vector conversion: resizes output if needed.
template<class T_out, class T_in>
void convert(hdnum::Vector<T_out>& out,
             const hdnum::Vector<T_in>& in)
{
    if (out.size() != in.size()) {
        out.resize(in.size());
    }

    for (std::size_t i = 0; i < in.size(); ++i) {
        out[i] = T_out(in[i]);
    }
}

// Convenience wrapper: allocates a new vector.
template<class T_out, class T_in>
hdnum::Vector<T_out> convert_vector(const hdnum::Vector<T_in>& in)
{
    hdnum::Vector<T_out> out(in.size());
    convert(out, in);
    return out;
}

// Matrix conversion: output matrix must already have correct shape.
template<class T_out, class T_in>

```

```

void convert(hdnum::DenseMatrix<T_out>& out,
             const hdnum::DenseMatrix<T_in>& in)
{
    if (out.rowsize() != in.rowsize() || out.colsize() != in.colsize()) {
        throw std::invalid_argument(
            "mpir::convert(DenseMatrix): output matrix has wrong size"
        );
    }

    for (std::size_t i = 0; i < in.rowsize(); ++i) {
        for (std::size_t j = 0; j < in.colsize(); ++j) {
            out[i][j] = T_out(in[i][j]);
        }
    }
}

// Convenience wrapper: allocates a new matrix.
template<class T_out, class T_in>
hdnum::DenseMatrix<T_out> convert_matrix(const hdnum::DenseMatrix<T_in>& in)
{
    hdnum::DenseMatrix<T_out> out(in.rowsize(), in.colsize());
    convert(out, in);
    return out;
}

```

Where the type conversion is done via the assignment:

```
out[i] = T_out(in[i])
```

for each element.

But this doesn't work for all pairs of `T_in`, `T_out` because not all the necessary constructors or conversion operators are implemented in the classes `CPFloat` and `FP`

Which Conversions Work which Don't ?

FP provides the constructors:

- `FP()`
- `FP(const double &)`
- `FP(const float &)`
- `FP(const int &)`

- `FP(const mpf_class &)`
- `FP<mm>(const FP<mm>&)`

So the conversion:

- `double ==> GMP`

works

`CPFloat` provides the constructors:

- `CPFloat()`
- `CPFloat(const double &)`
- `CPFloat<mm, ee>(const double &)`

So:

- `double ==> CPFloat`

works

`CPFloat` also provides the conversion operator `operator double() const`. Therefore

- `CPFloat ==> double`

works as well.

Because `CPFloat` can be converted to `double` and `GMP` can be constructed from a `double`:

- `CPFloat ==> GMP`

works too, by the implicit conversion of `CPFloat` to a `double`

But since the `FP` class doesn't provide a `double()` operator the conversion

- `GMP ==> double`

doesn't work.

Also `CPFloat` doesn't implement a constructor that accepts a `GMP` type, (or also because `FP` doesn't provide a `double()` operator) the conversion:

- `GMP ==> CPFloat`

doesn't work.

We summarize this with the following diagram and table:

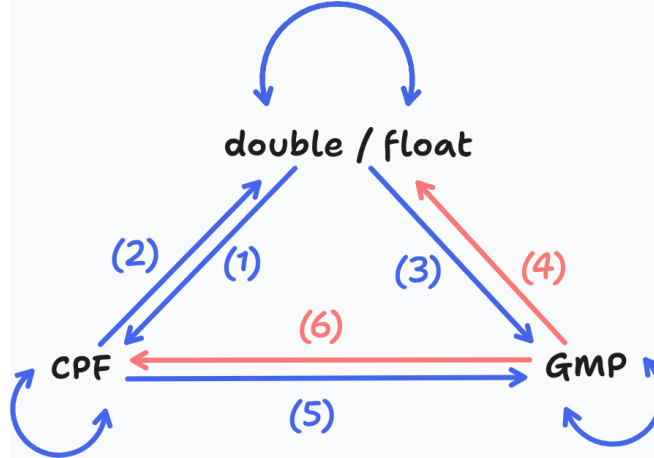


Figure 3.1: conversion triangle

Table 3.5: conversion table

Nr	Conversion	Works	
		?	Why ?
(1)	<code>double</code> $\Rightarrow$ <code>CPFloat</code>	Yes	<code>CPFloat(const double &)</code>
(2)	<code>CPFloat</code> $\Rightarrow$ <code>double</code>	Yes	<code>CPFloat::double()</code>
(3)	<code>double</code> $\Rightarrow$ <code>GMP</code>	Yes	<code>FP(const double &)</code>
(4)	<code>GMP</code> $\Rightarrow$ <code>double</code>	No	No <code>FP::double()</code>
(5)	<code>CPFloat</code> $\Rightarrow$ <code>GMP</code>	Yes	<code>CPFloat</code> is converted to <code>double</code> implicitly
(6)	<code>GMP</code> $\Rightarrow$ <code>CPFloat</code>	No	No <code>FP(const CPFloat &)</code> or no <code>FP::double()</code>

So if the conversion operator `FP::double()` would be implemented both (4) and (6) would work. (6) would work too because `GMP` type would be implicitly converted to a `double` and a `CPFloat` would be constructed. But more explicitly a constructor `CPFloat(const FP &)` can be implemented too.

But since we are initially not allowed to modify `HDNUM` source code we take a different approach.

3.2.2 Structure of FP and CPFloat Classes

FP	CPFloat
-mpf_class number	-double number
+FP() +FP(const double &) +FP(const mpf_class &) +FP(...) +getNumber() : mpf_class	+CPFloat() +CPFloat(const double &) +CPFloat(...) +operator double() +getNumber() : double

- FP:
 - FP has a single private member variable `number` that is an GMP type, i.e. `mpf_class`.
 - it can be accessed with the public getter `const getNumber() : mpf_class`
- CPFloat:
 - CPFloat has a single private member variable `number` of type `double`.
 - it can be accessed with the public getter `const getNumber() : double`

Note

There's a mistake in HDNUM `highprec_gmp.hh` source file. The member variable `number` is documented as:

Listing 3.1 `highprec_gmp.hh`

```
private:  
    // stores a double to represent the low precision number  
    mpf_class number;
```

The comment `// stores a double to represent the low precision number` is wrong, and is probably copied over by mistake from `lowprec_cpfloat.hh`:

Listing 3.2 lowprec_cpfloat.hh

```
private:
    // stores a double to represent the low precision number
    double number;
```

Instead it should be something like:

```
// stores an mpf_class to represent the high precision GMP number
```

With the above interface, given a GMP number, the double representation of the number can be obtained with:

Listing 3.3 get_d() method

```
hdnum::FP128 fp128_number(1.2345);
auto d = fp128_number.getNumber().get_d(); //d is double
```

Where we use the `.get_d()` method of `mpf_class` to get the `double` representation of the GMP number.

3.2.3 Implementing Conversions - Correct Approach

The main issue is that

```
out[i] = T_out(in[i]);
```

does not work uniformly for all type pairs. We replace this direct construction step with:

```
out[i] = scalar_convert<T_out>(in[i]);
```

This selects the appropriate conversion path at compile time, depending on the input and output types using standard library type traits and `if constexpr`

As summarized in Figure 3.1 and Table 3.5, the only conversions that do not work by direct construction are:

1. FP $\Rightarrow$ double,
2. FP $\Rightarrow$ CPFloat

All other relevant cases can be handled by direct construction of the form:

```
T_out(x) // where x has type T_in
```

In C++, we can check at compile time whether an object of type `T_out` can be constructed from an argument type `const T_in&` using the standard library type trait:

```
std::is_constructible_v<T_out, const T_in&>
```

This expression evaluates to `true` if such a construction is valid, and to `false` otherwise.

It can be verified that this indeed evaluates to `false` only for the conversion pairs:

- `T_out == double, T_in == FP`
- `T_out == CPFloat, T_in == FP`

Then, the initial part of `scalar_cast` can be written as:

```
template<class T_out, class T_in>
T_out scalar_cast(const T_in& x)
{
    if constexpr (std::is_constructible_v<T_out, const T_in&>) {
        return T_out(x); // we use the direct construction
    }
    else { //remaining cases: FP => double || FP => CPFloat
        ...
    }
}
```

Logically, we know that the only possible cases in the `else` branch are:

- `T_out == double, T_in == FP`
- `T_out == CPFloat, T_in == FP`

In both cases `T_in` is an FP type. as we've seen in Listing 3.3 the `double` representation of an FP number `x` can be extracted with:

```
x.getNumber().get_d()
```

Since both `CPFloat` and `double` can be constructed from a `double`, we could simply write:

```

template<class T_out, class T_in>
T_out scalar_cast(const T_in& x)
{
    if constexpr (std::is_constructible_v<T_out, const T_in&>) {
        return T_out(x); // we use the direct construction
    }
    else { //remaining cases: FP => double || FP => CPFloat
        return T_out(x.getNumber().get_d());
    }
}

```

However, this version is not robust enough because the `else` branch relies on our prior reasoning that only the two exceptional cases can reach it. The code itself does not express this assumption. If `scalar_cast` is later instantiated with another unsupported conversion pair, the compiler will still try to compile

```
x.getNumber().get_d()
```

even if `T_in` is not an FP type. This should lead to implementation-dependent and potentially confusing error message.

Therefore, it is better to make the remaining cases explicit using another `if constexpr` condition:

```

template<class T_out, class T_in>
T_out scalar_cast(const T_in& x)
{
    if constexpr (std::is_constructible_v<T_out, const T_in&>) {
        return T_out(x); // direct construction works
    }
    else if constexpr (
        is_hdnum_fp_v<T_in> &&
        std::is_constructible_v<T_out, double>
    ) {
        return T_out(x.getNumber().get_d());
    }
    else {
        static_assert(always_false<T_out, T_in>::value,
            "unsupported scalar conversion");
    }
}

```

Here, the helper trait `is_hdnum_fp_v<T>` detects whether a type `T` is one of HDNUM's GMP-based high precision number types `hdnum::FP<m>`. This is a compile-time Boolean value, and analogous to the standard library traits such as `std::is_constructible_v`.

The implementation, given below, follows a common template metaprogramming partial specialization pattern:

```
// Detect HDNUM's GMP wrapper type hdnum::FP<m>
template<class T>
struct is_hdnum_fp_impl : std::false_type {};

template<int m>
struct is_hdnum_fp_impl<hdnum::FP<m>> : std::true_type {};

template<class T>
inline constexpr bool is_hdnum_fp_v =
    is_hdnum_fp_impl<
        std::remove_cv_t<std::remove_reference_t<T>>
    >::value;
```

1) The first part

```
template<class T>
struct is_hdnum_fp_impl : std::false_type {};
```

is the default case and says:

for an arbitrary type `T`, it is false

So:

```
is_hdnum_fp_impl<double>::value      // false
is_hdnum_fp_impl<float>::value        // false
is_hdnum_fp_impl<hdnum::FP16>::value // false, because CPFloat, not GMP FP
```

2) The second part

```
template<int m>
struct is_hdnum_fp_impl<hdnum::FP<m>> : std::true_type {};
```

is the specialization case, and says:

if the type has the form `hdnum::FP<m>`, it is true.

So:

```
is_hdnum_fp_impl<hdnum::FP128>::value // true
is_hdnum_fp_impl<hdnum::FP256>::value // true
```

3) The last part

```
template<class T>
inline constexpr bool is_hdnum_fp_v =
    is_hdnum_fp_impl<
        std::remove_cv_t<std::remove_reference_t<T>>
    >::value;
```

defines a compile-time boolean variable template. For example:

```
is_hdnum_fp_v<double> // false
is_hdnum_fp_v<hdnum::FP128> // true
```

It is analogous to standard library helpers such as:

```
std::is_arithmetic_v<T>
std::is_constructible_v<T, Args...>
std::is_convertible_v<From, To>
```

and allows to write

```
is_hdnum_fp_v<T>
```

instead of

```
is_hdnum_fp_impl<T>::value
```

The keyword `inline` is needed for variable that are defined in header files.

The use of:

```
std::remove_reference_t<T>
std::remove_cv_t<T>
```

makes the trait insensitive to references and `const` / `volatile` qualifiers. For example this way all of the following are treated as the same:

```
hdnum::FP<64>
const hdnum::FP<64>
hdnum::FP<64>&
const hdnum::FP<64>&
```

Without removing references and cv-qualifiers, the specialization would not match types such as `hdnum::FP<m>&`

The final branch of `scalar_cast`

```
else{
    static_assert(always_false<T_out, T_in>::value,
        "unsupported scalar conversion");
}
```

This branch is reached only if neither of the previous compile-time conditions applies:

```
std::is_constructible_v<T_out, const T_in&>
```

is `false`, so direct construction does not work, and

```
is_hdnum_fp_v<T_in> && std::is_constructible_v<T_out, double>
```

is also `false`, so the special `FP => double / FP => CPFloat` conversion path doesn't apply either.

Therefore, if the compiler reaches this final 'else' branch, the requested scalar conversion is unsupported. In that case, we want compilation to fail with a clear error message.

The helper trait is defined as:

```
template<class...>
struct always_false : std::false_type {};
```

This defines a type trait that is always `false`, regardless of the template arguments. For example:

```
always_false<int>::value           // false
always_false<double, int>::value   // false
always_false<hdnum::FP<64>, double>::value // false
```

The `class...` syntax denotes a template parameter pack and means that `always_false` can accept any number of template type arguments. In our case we use two arguments:

```
always_false<T_out, T_in>::value
```

It is important that `always_false<T_out, T_in>::value` depends on template parameters `T_out` and `T_in`, as it lets us create a compile-time `false` value that is evaluated when the templates are instantiated and not when the function is parsed. Omitting and writing simply `static_assert(false, ...)` would cause the assertion to be checked when the function template is parsed, and not only when the template is instantiated and the final branch is actually selected for the concrete instantiation.

scalar_cast Final Version

```
#pragma once

#include <cstdint>
#include <stdexcept>
#include <type_traits>

#include "hnum.hh"

namespace mpir {

// Helper for dependent static_assert(false)
template<class...>
struct always_false : std::false_type {};

// Detect HDNUM's GMP wrapper type hnum::FP<m>
template<class T>
struct is_hnum_fp_impl : std::false_type {};

template<int m>
struct is_hnum_fp_impl<hnum::FP<m>> : std::true_type {};

template<class T>
inline constexpr bool is_hnum_fp_v =
    is_hnum_fp_impl<
        std::remove_cv_t<std::remove_reference_t<T>>
    >::value;

// Scalar conversion
template<class T_out, class T_in>
T_out scalar_cast(const T_in& x)
{
    if constexpr (std::is_constructible_v<T_out, const T_in&>) {
        return T_out(x);
    }
    else if constexpr (
        is_hnum_fp_v<T_in> &&
        std::is_constructible_v<T_out, double>
    ) {

```



```

        return T_out(x.getNumber().get_d());
    }
    else {
        static_assert(always_false<T_out, T_in>::value,
            "mpir::scalar_cast: unsupported scalar conversion");
    }
}

// Vector conversion
template<class T_out, class T_in>
void convert(hdnum::Vector<T_out>& out,
             const hdnum::Vector<T_in>& in)
{
    if (out.size() != in.size()) {
        throw std::invalid_argument(
            "mpir::convert(Vector): output vector has wrong size"
        );
    }

    for (std::size_t i = 0; i < in.size(); ++i) {
        out[i] = scalar_cast<T_out>(in[i]);
    }
}

// DenseMatrix conversion
template<class T_out, class T_in>
void convert(hdnum::DenseMatrix<T_out>& out,
             const hdnum::DenseMatrix<T_in>& in)
{
    if (out.rowsize() != in.rowsize() || out.colsize() != in.colsize()) {
        throw std::invalid_argument(
            "mpir::convert(DenseMatrix): output matrix has wrong size"
        );
    }

    for (std::size_t i = 0; i < in.rowsize(); ++i) {
        for (std::size_t j = 0; j < in.colsize(); ++j) {
            out[i][j] = scalar_cast<T_out>(in[i][j]);
        }
    }
}

```

```

    }
}

// convenience function
template<class T_out, class T_in>
hdnum::DenseMatrix<T_out> convert_matrix(const hdnum::DenseMatrix<T_in>& in)
{
    hdnum::DenseMatrix<T_out> out(in.rowsize(), in.colsize());
    convert(out, in);
    return out;
}

// convenience function
template<class T_out, class T_in>
hdnum::Vector<T_out> convert_vector(const hdnum::Vector<T_in>& in)
{
    hdnum::Vector<T_out> out(in.size());
    convert(out, in);
    return out;
}

} // namespace mpir

```

3.3 Testing `hdnum_conversions.hpp`

The file `test_conversion.cc` tests the conversion infrastructure implemented in `hdnum_conversions.hpp`. Its goals are to test that:

1. `scalar_cast<T_out>(x)` compiles for representative type.
2. `scalar_cast<T_out>(x)` returns the requested output type.
3. lossy low-precision conversions visibly lose information.
4. `convert(Vector)` applies `scalar_cast` element-wise correctly.
5. `convert(DenseMatrix)` applies `scalar_cast` elementwise correctly.
6. Vector and Matrix sizes are handled correctly.
7. Wrong-sized output containers throw exceptions
8. The convenience functions `convert_vector` and `convert_matrix` allocate correctly.

3.3.1 Scalar Conversion Tests

The scalar tests check that

```
mpir::scalar_cast<T_out>(x)
```

can be instantiated for representative input/output type pairs. For each pair, several sample values are tested:

```
0.0  
1.0  
-2.5  
1.0 / 3.0  
1.23456789012345
```

The result is checked at compile time with

```
static_assert(std::is_same_v<decltype(y), T_out>);
```

To verify that `scalar_cast<T_out>(x)` returns the requested output type `T_out`.

Both the input and the converted value are transformed to their approximate `double` representation with `approx_as_double` in order to compare the values with various `assert` and inspect the outputs.

3.3.2 Lossy Conversion Tests

It is verified that conversion from `double` to `CPFloat` indeed loses precision using a value such as 1.23456789012345.

This is only a useful sanity check that the conversion is really taking place.

3.3.3 Lossless Conversion Tests

It is also verified that conversion from a lower precision to a higher precision do not indeed lose information.

3.3.4 Vector Conversion Tests

Vector tests check the function

```
mpir::convert(hdnum::Vector<T_out>& out,  
              const hdnum::Vector<T_in>& in)
```

for a representative vector of length 5 and checks that each component of the output agrees with applying `scalar_cast<T_out>` to the corresponding input component.

The comparison is performed approximately by converting both values to `double` and using a small relative tolerance with the function

```
inline bool close(double a, double b, double atol = 0.0, double rtol = 1e-12)  
{  
    const double scale = std::max(std::abs(a), std::abs(b));  
    return std::abs(a - b) <= atol + rtol * scale;  
}
```

This function says whether the two double values are close enough to be considered equal. The tolerance is scaled by the max value of the inputs.

The test also includes the edge cases:

- vector size 0
- vector size 1

Additionally wrong-sized output vector is checked to verify whether an exception is thrown.

Finally the convenience function

```
mpir::convert_vector<T_out>(in)
```

is tested, by verifying the resulting type and converted values.

3.3.5 Matrix Conversion Tests

The matrix tests check the function

```
mpir::convert(hdnum::DenseMatrix<T_out>& out,  
              const hdnum::DenseMatrix<T_in>& in)
```

on a representative 5 x 5 matrix, by looping over the output values and comparing them against the result of applying `scalar_cast` to corresponding input entries.

The test includes the edge case of 1 x 1 matrix and a wrong-sized test.

The convenience function `mpir::convert_matrix<T_out>(in)` is also tested.

3.3.6 Representative Pair Types

1. Native $\leftrightarrow$ Native:
 - FP32 $\leftrightarrow$ FP64
2. Working precision $u \rightarrow$ low precision u_f :
 - FP64 $\rightarrow$ FP16 / bfloat16 / FP8
3. Low Precision $u_f \rightarrow$ working precision u :
 - FP16 / bfloat16 / FP8 $\rightarrow$ FP64
4. Working precision $u \rightarrow$ residual precision u_r
 - FP64 $\rightarrow$ FP128 / FP256
5. Residual precision $u_r \rightarrow$ working precision u
 - FP128 / FP256 $\rightarrow$ FP64
6. Low precision $u_f \leftrightarrow$ high residual/reference precision u_r
 - FP128 / FP256 $\leftrightarrow$ FP16 / bfloat16 / FP8
7. High precision $\leftrightarrow$ high precision
 - FP128 $\leftrightarrow$ FP256

Covering the main conversion direction required later by the algorithm:

- $u \rightarrow u_f$
- $u_f \rightarrow u$
- $u \rightarrow u_r$
- $u_r \rightarrow u$

The tests also include low to high-precision conversions not required directly by the algorithm but useful nevertheless.

4 Report

4.1 This is some title